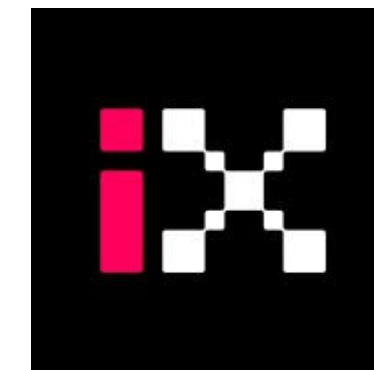
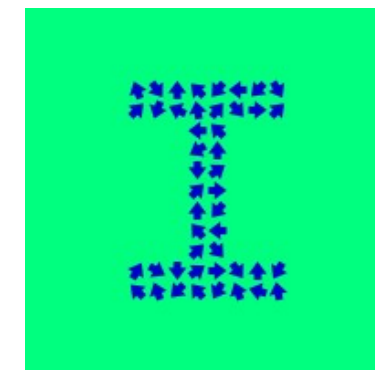


Emulating the REACH dipole beam pattern

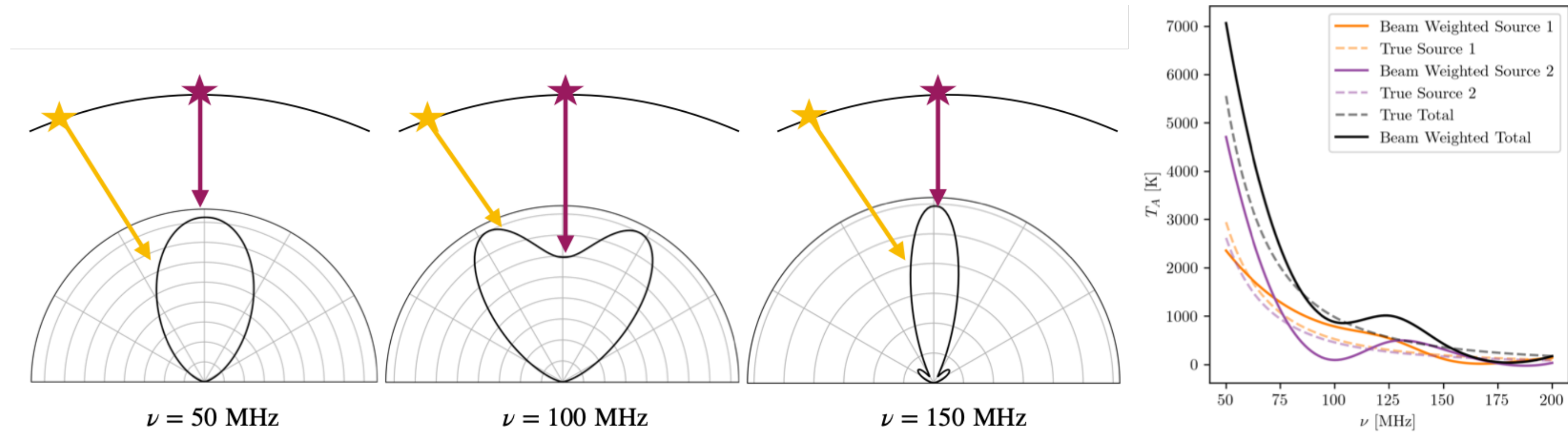
Harry Bevins

With John Cumner, Dominic Anstey, Jean Cavillot, Jacob Tutt

IMPERIAL

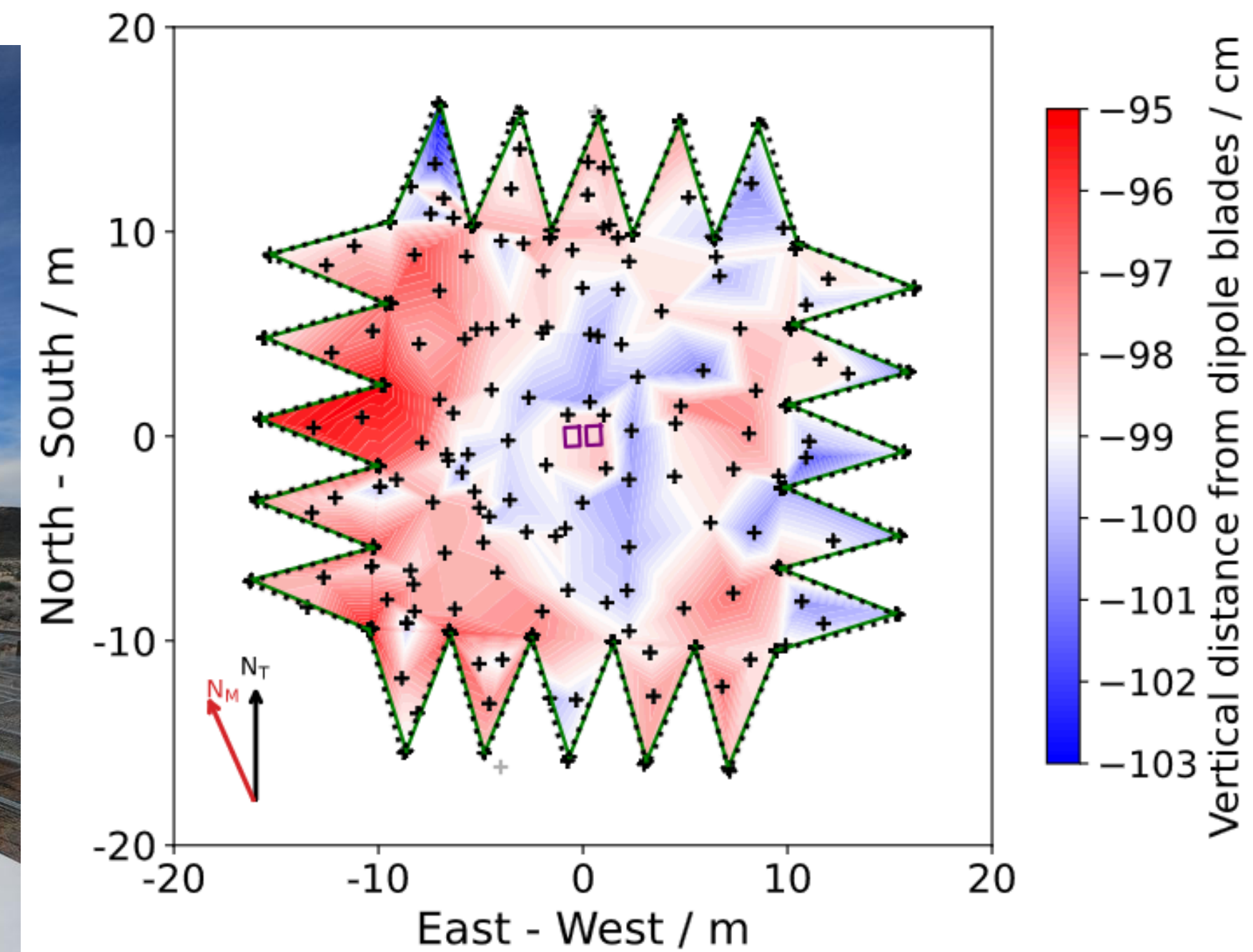


The problem...



- Up until recently we used a fixed beam in the analysis pipeline
- Cumner et al. 2023 and Pattison et al. 2026 showed that this might not be good enough...

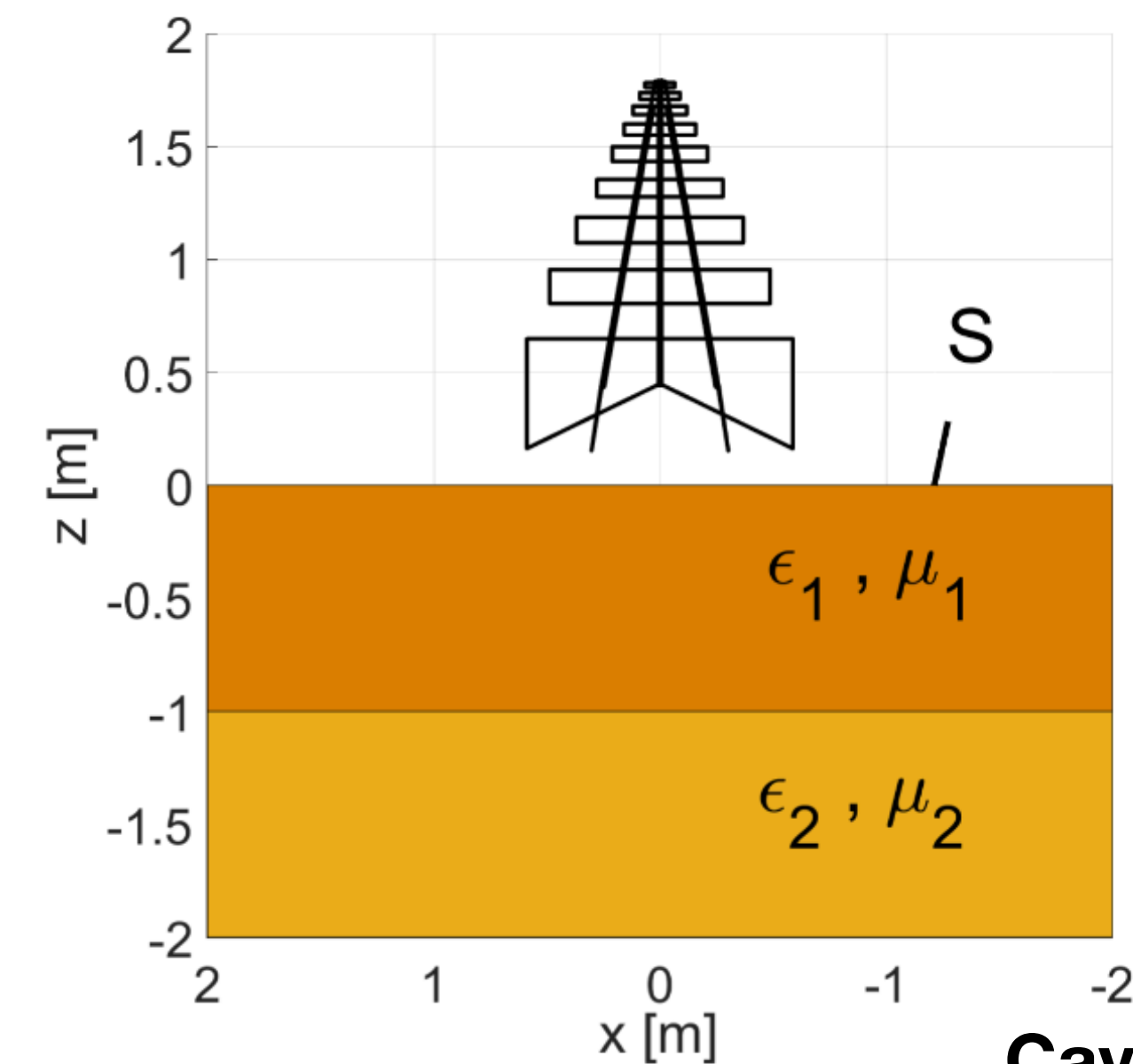
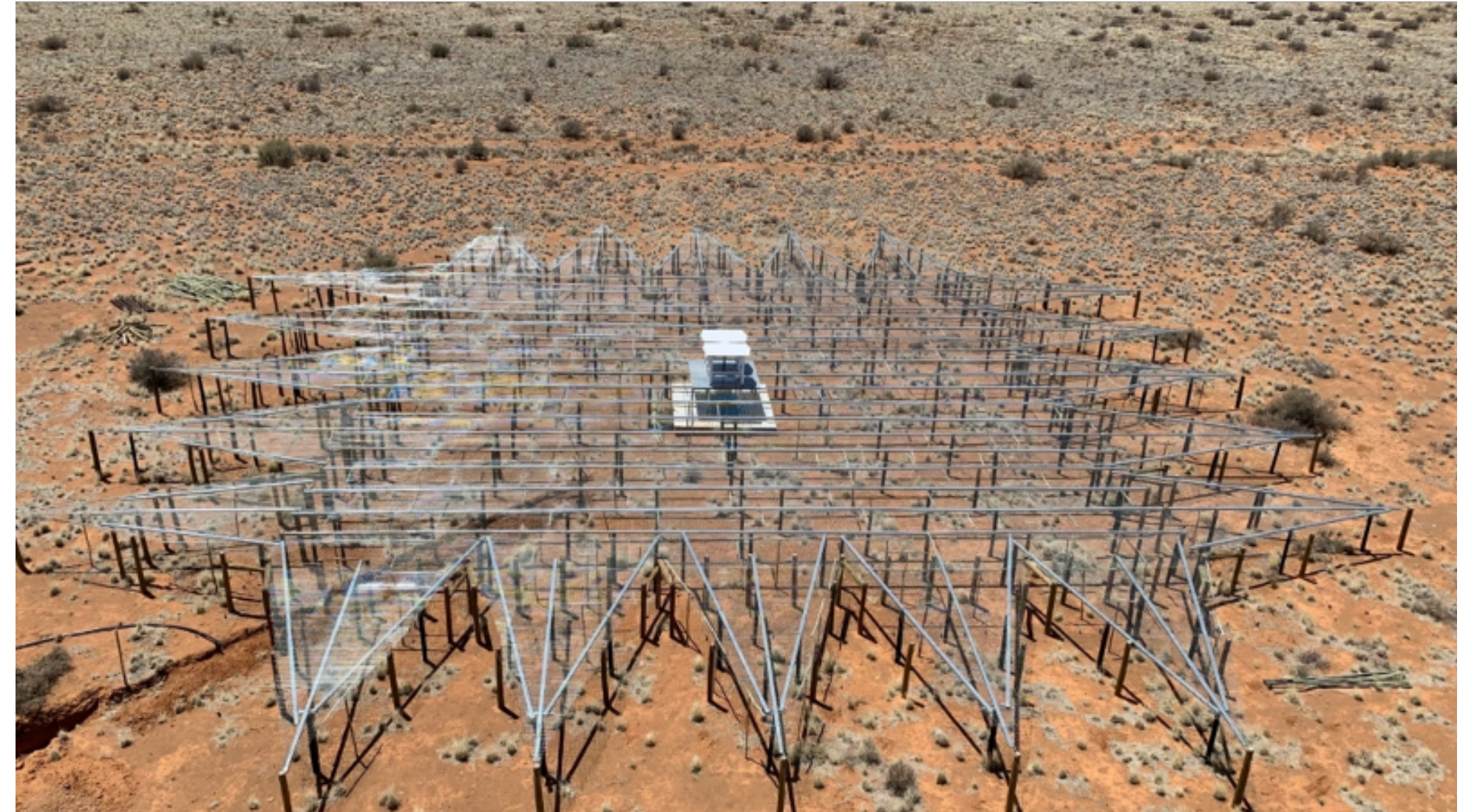
The problem...



Pattison + Cumner et al. 2026

Lets fit for the beam...

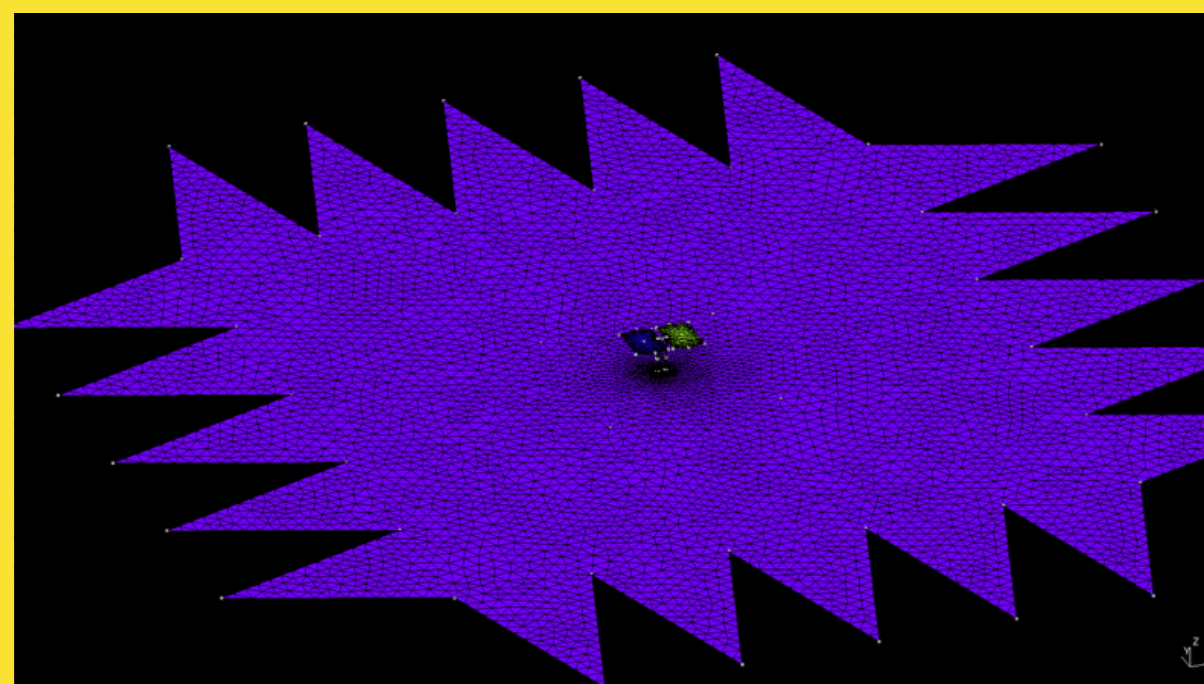
- Ideally we want to fit for
 - the geometry of the antenna
 - And the environment
- ... alongside the foregrounds, signal etc...
- CST is very expensive to run and hard to include soil
- Use Jean Cavillot's MoM simulation suit and build surrogate models...



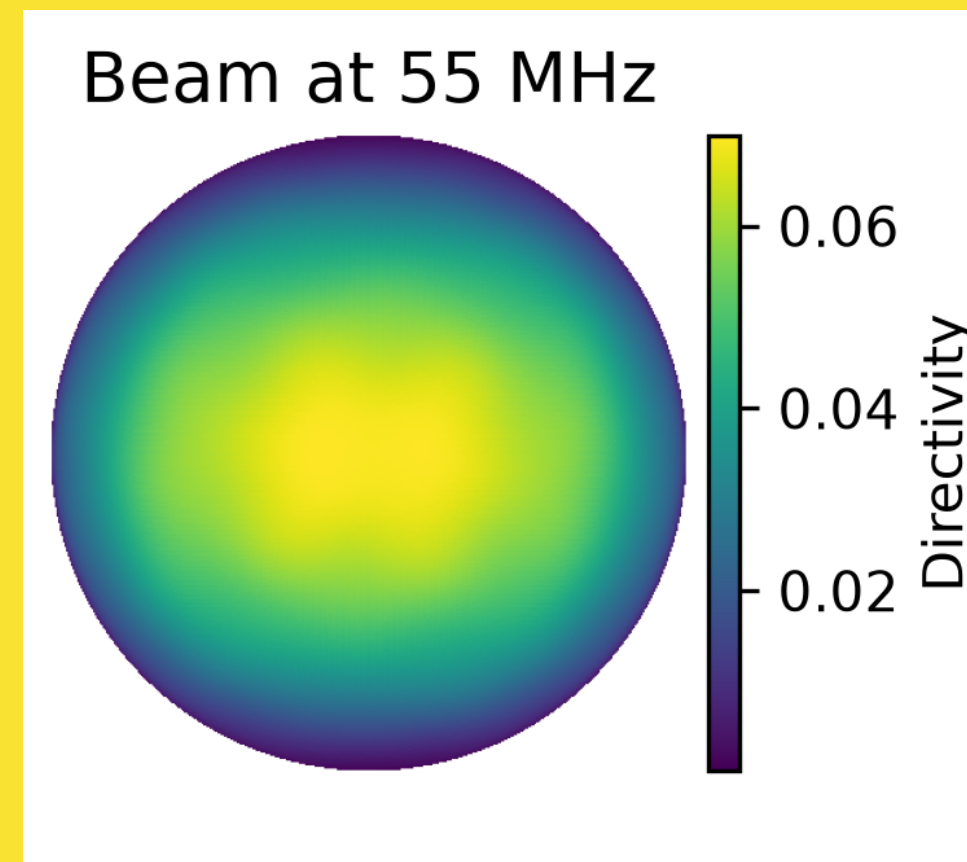
Cavillot et al. 2024



GMSH



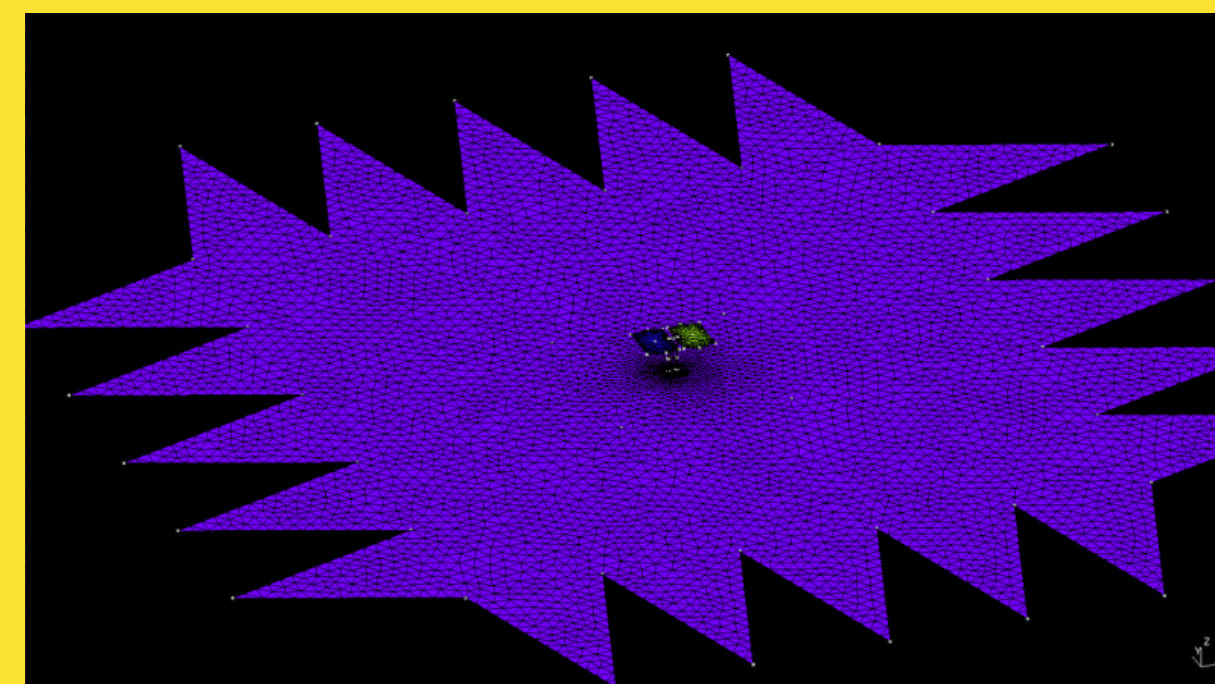
MoM



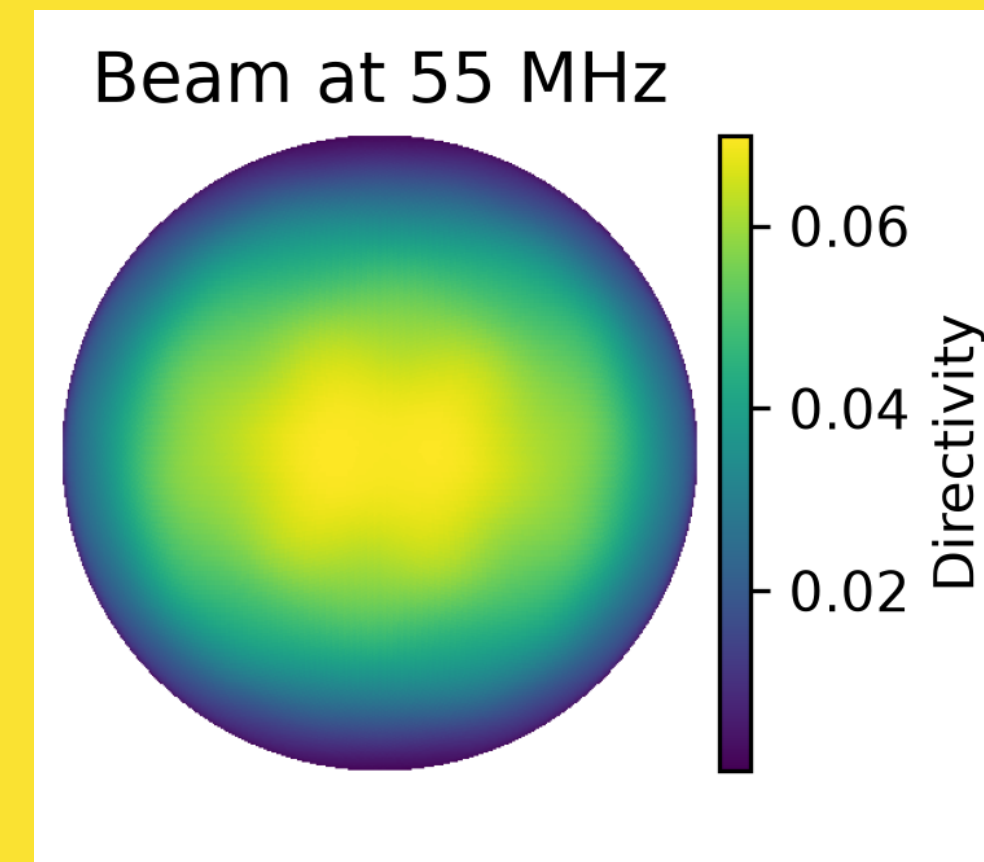
1500 Simulations of REACH-like dipole antennas



GMSH

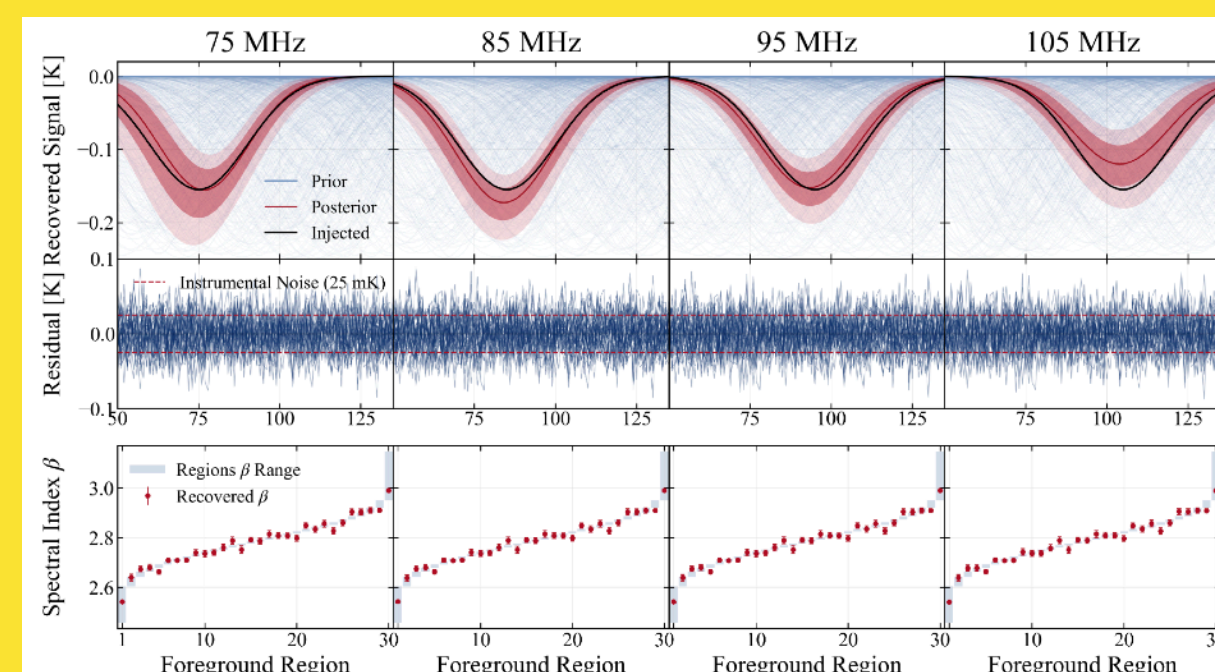


MoM

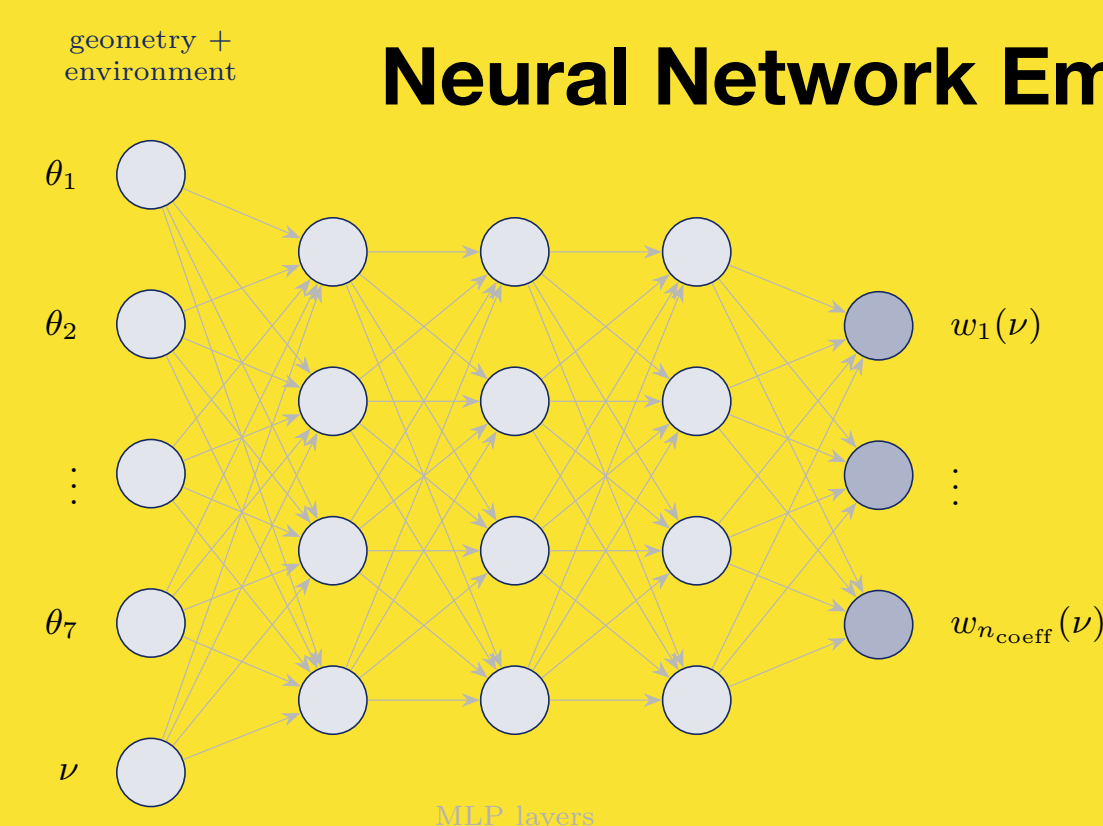


1500 Simulations of REACH-like dipole antennas

Analytic Marginalisation - Jacob's Talk

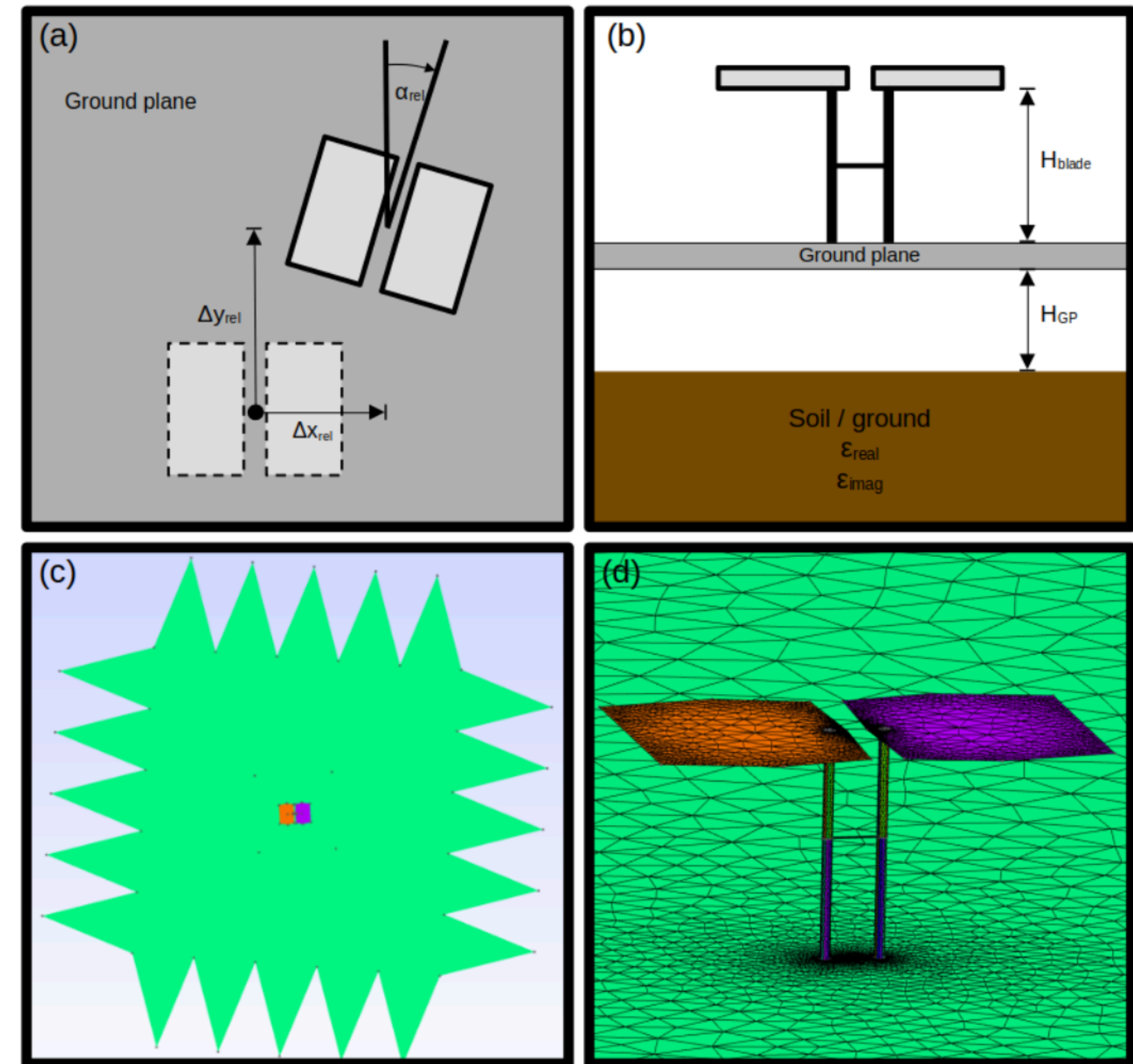


Neural Network Emulation



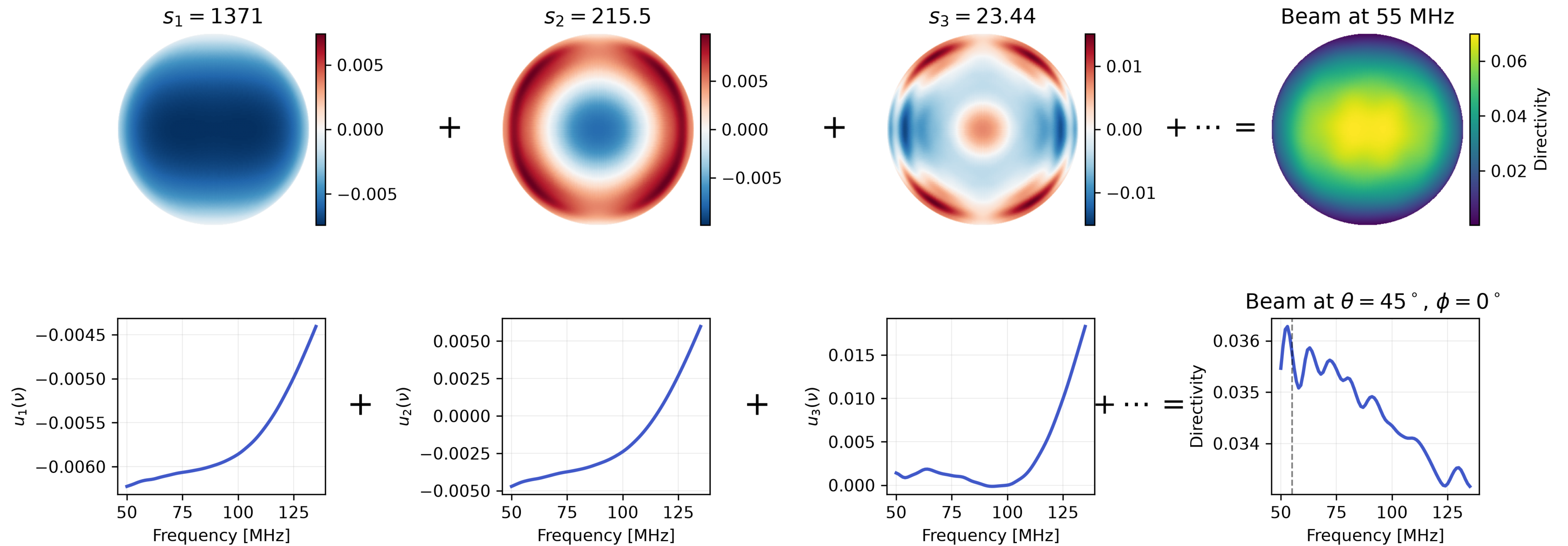
What parameters are we varying?

- Height of the ground plane
- Height of the blades
- Rotation and shift of the blades
- Soil permittivity
- If you think we should be varying something else let us know...



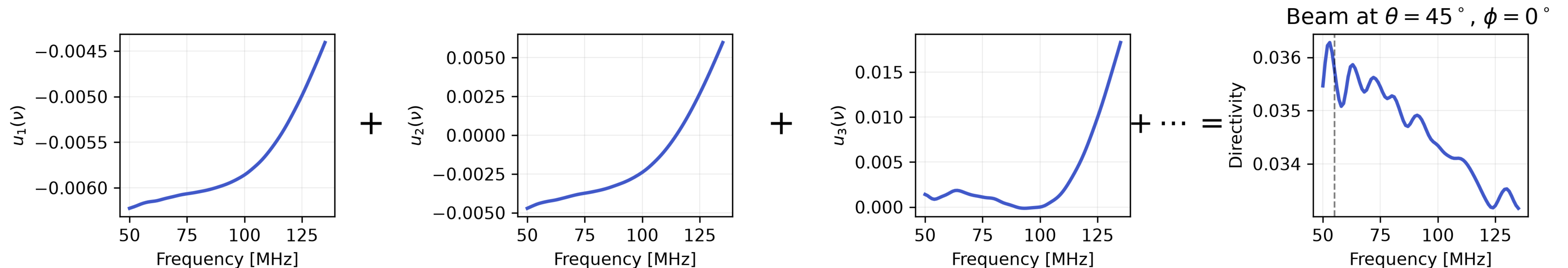
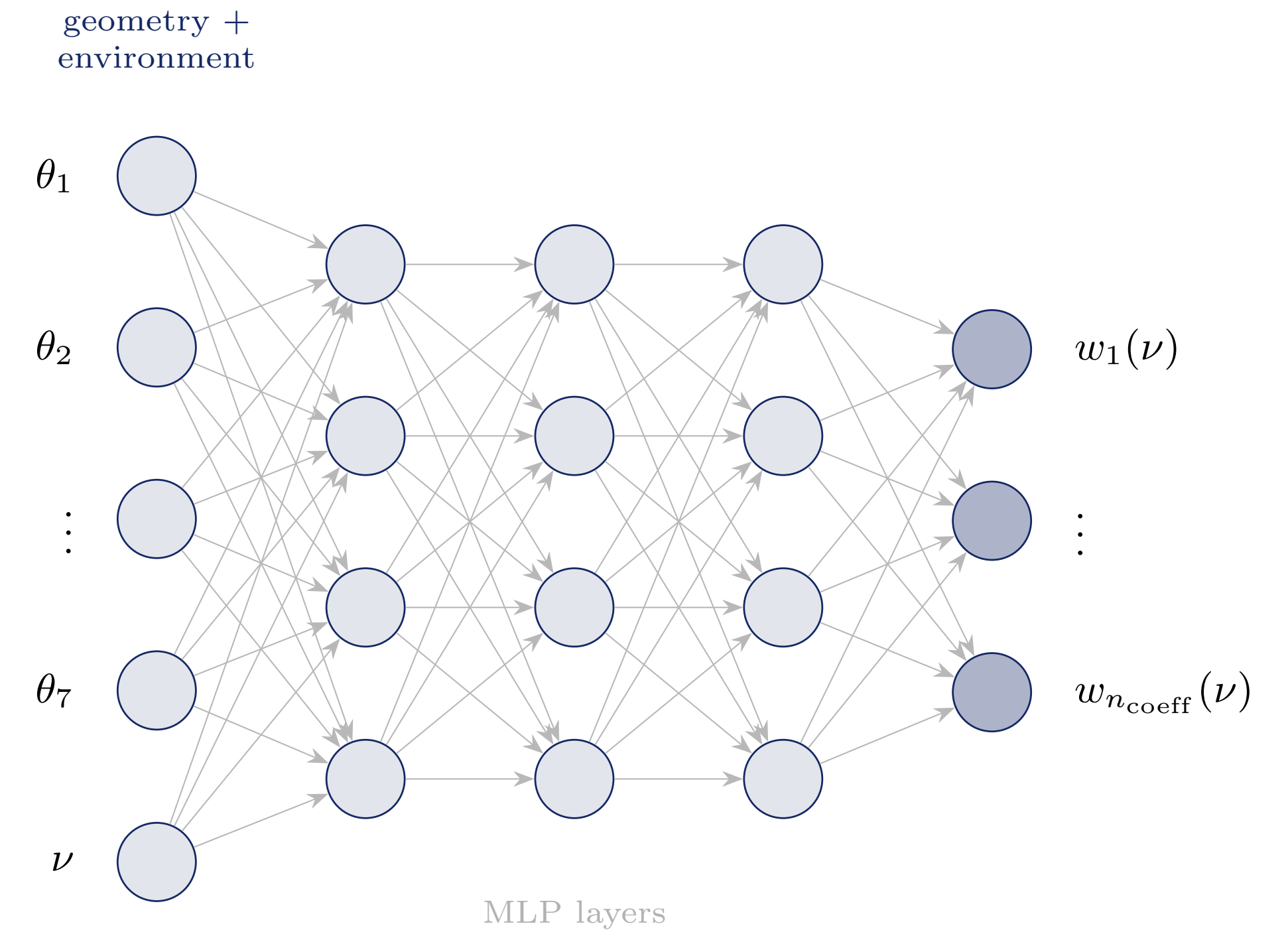
Tutt et al. 2026

From directivity to SVD...



The setup...

- Using a boring MLP
- Takes in geometry + environment parameters + frequency
- Predicts N_{coeff} weights $w_i(\nu_j)$



Assessing Emulator Accuracy

- How does an error in the weights propagate through to the residuals after foreground simulation

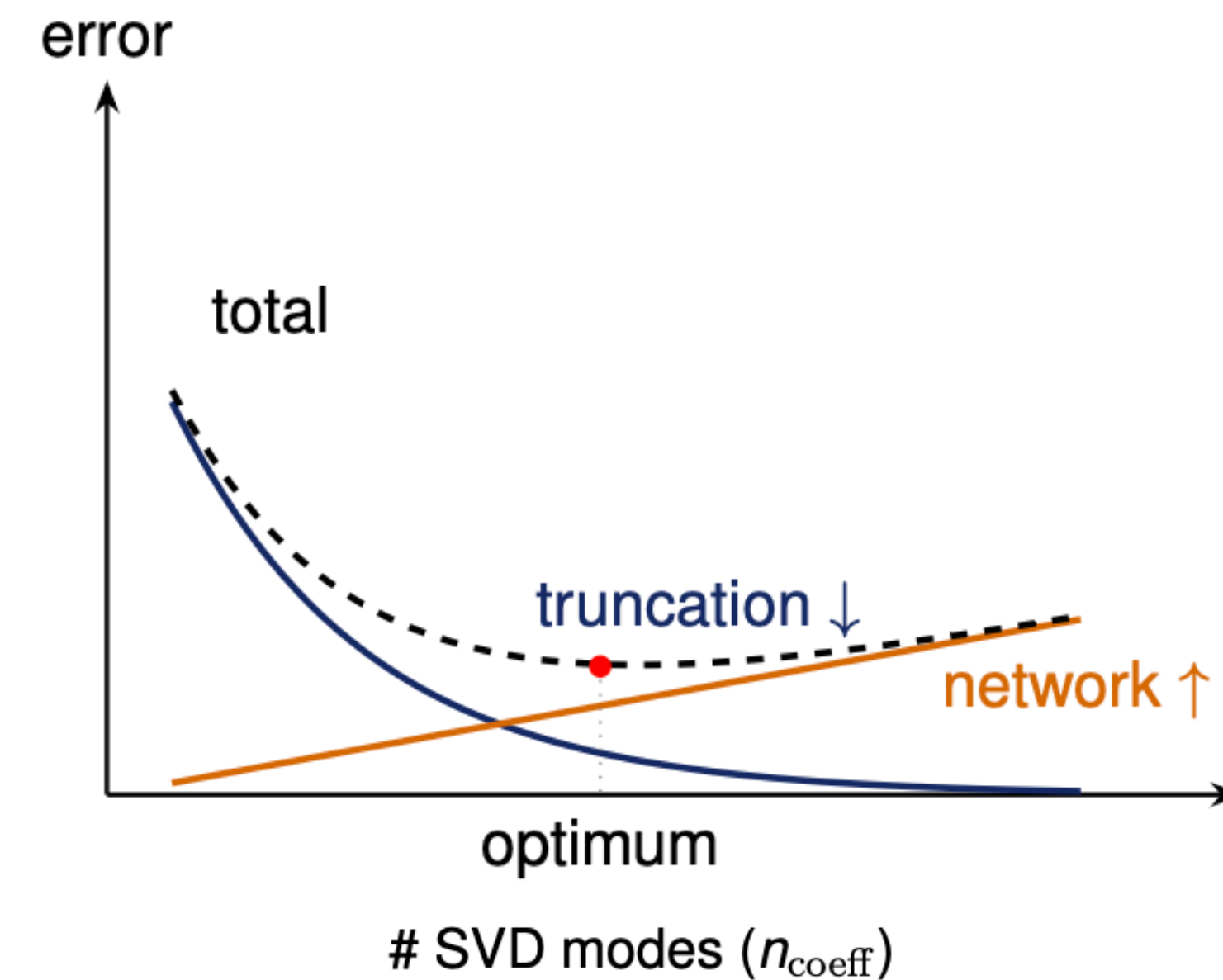
$$J_k(\nu) = \frac{\delta T_{ant}}{\delta w_k}$$

$$L = \frac{1}{N} \sum_k (\sigma_k^{FG})^\beta (w_k(\nu) - f(\theta, \nu))^2$$

- Assuming GSM sky map and uniform scaling across the sky

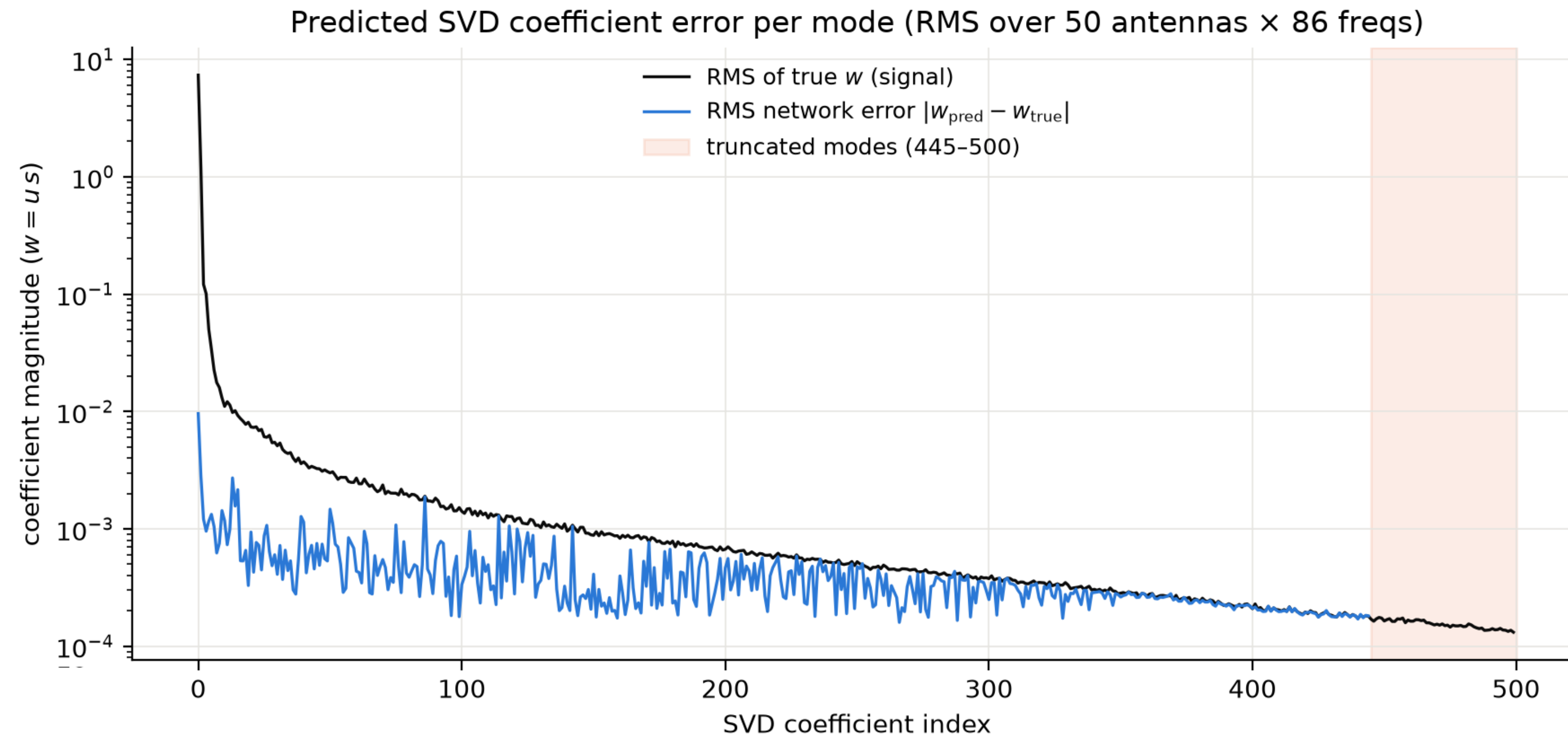
Tuning the network

- Automatic search through some prior space on the hyper parameters of the problem
- Tuning the architecture of the network
- The number of predicted coefficients
- The weighting in the loss function

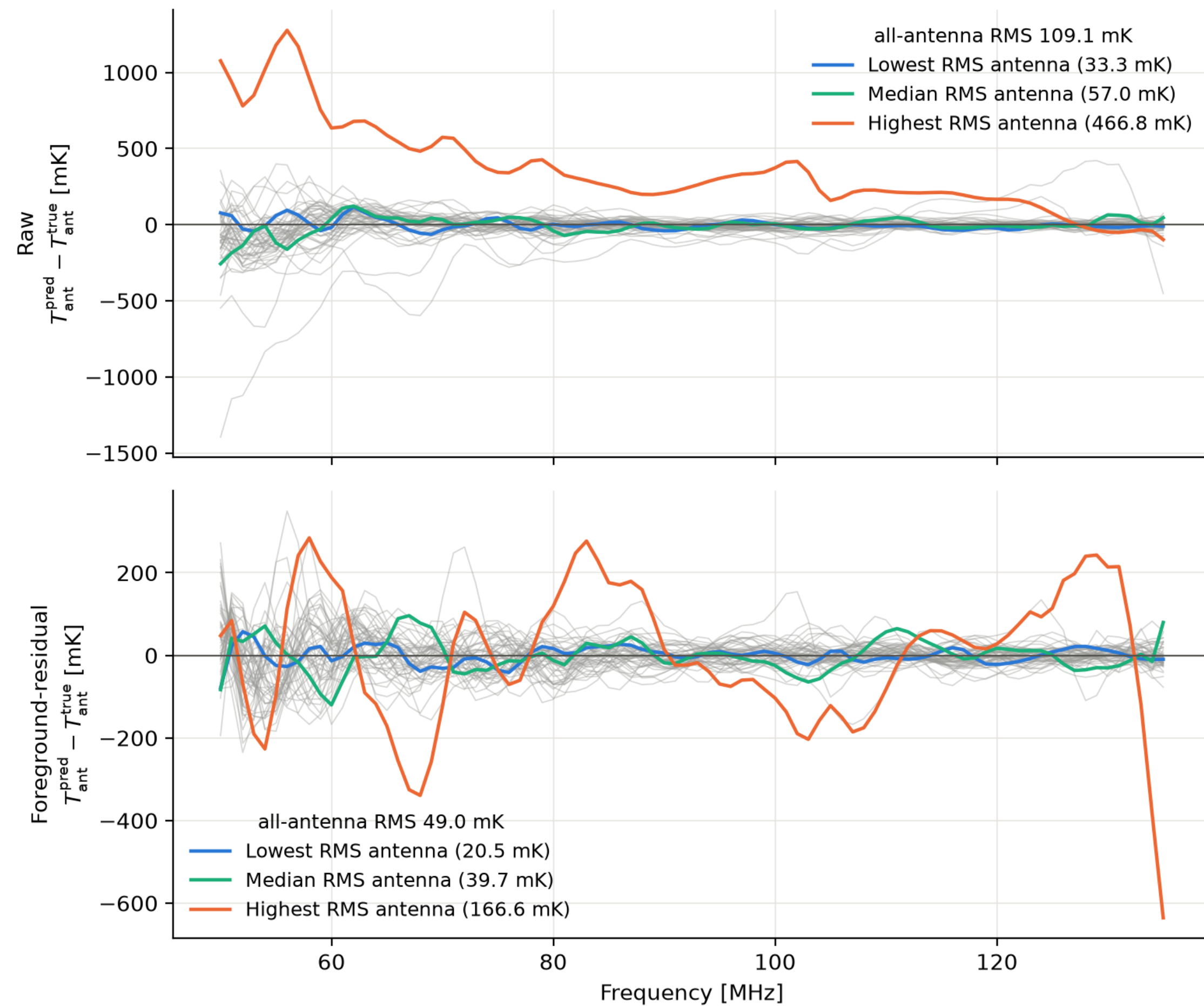


Network Performance

Averaged over antenna and frequency
as a function of **SVD coeff**



Network Performance



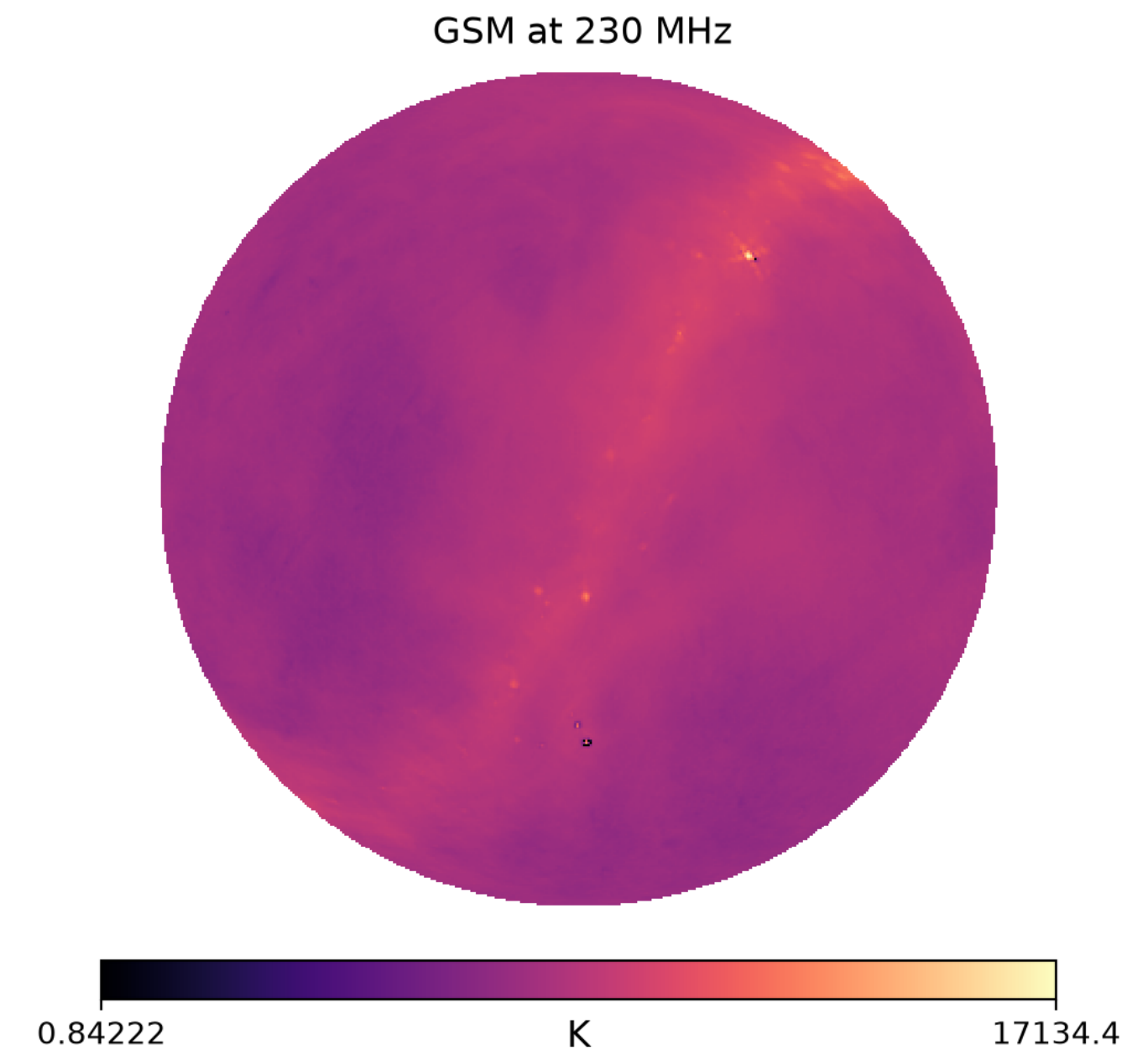
GSM uniformly scaled over frequency
and convolved with the beam

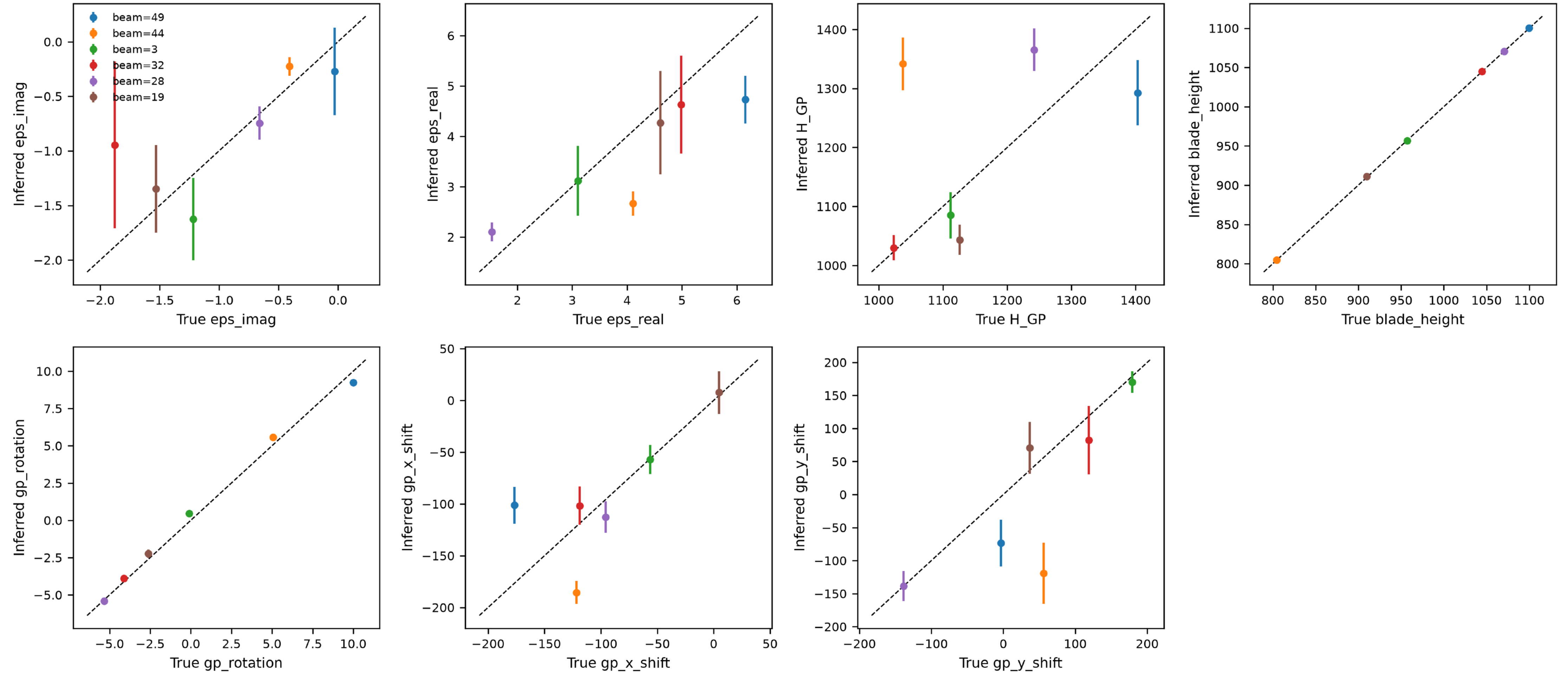
Effectively how much of the uncertainty
from the beam emulation creeps into the
residuals after a log polynomial
foreground fit

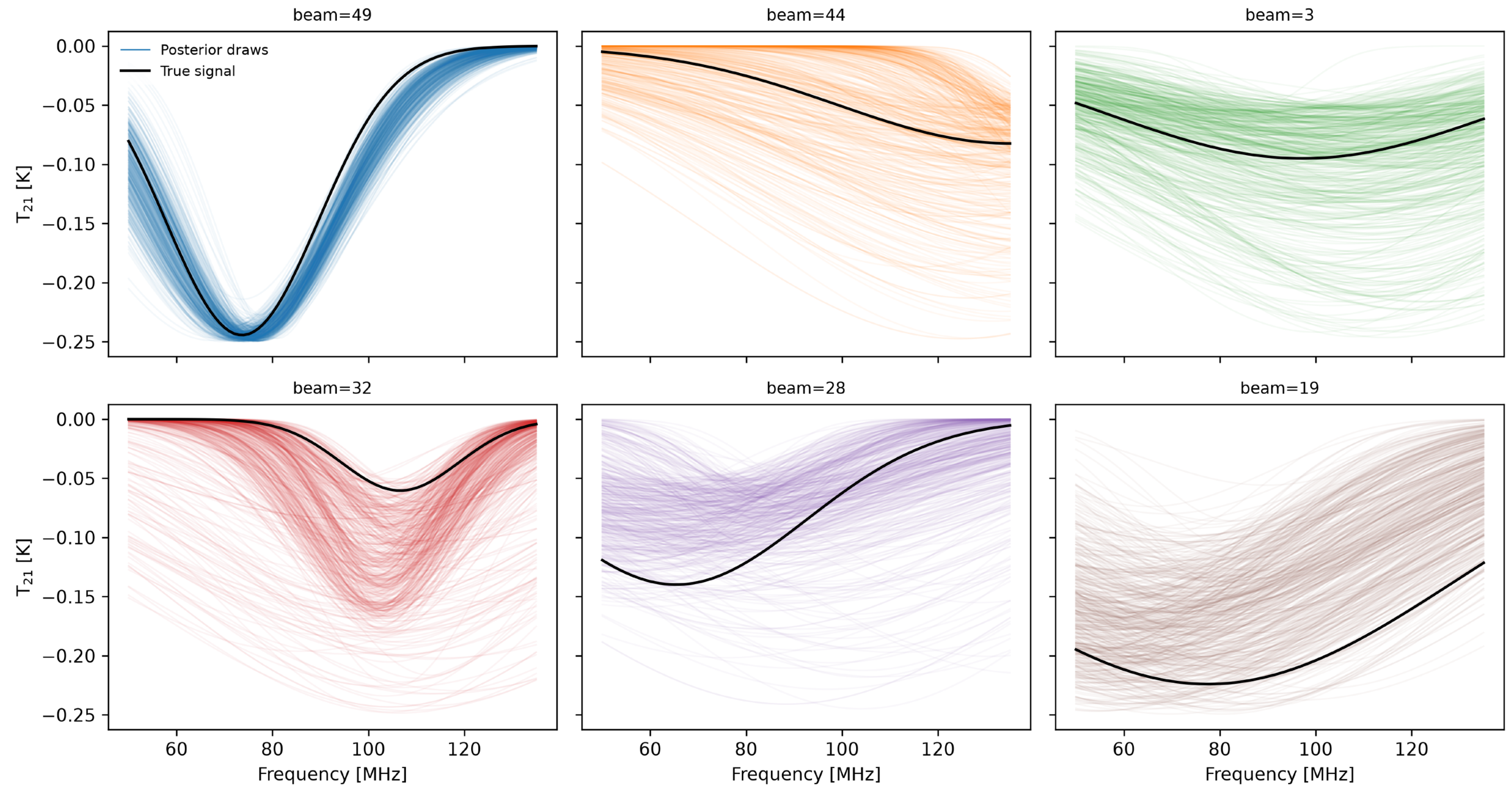


Mock Inference

- Again this is just GSM uniformly scaled across frequency
- Plus a random gaussian signal profile
- And a random beam from a held out test set

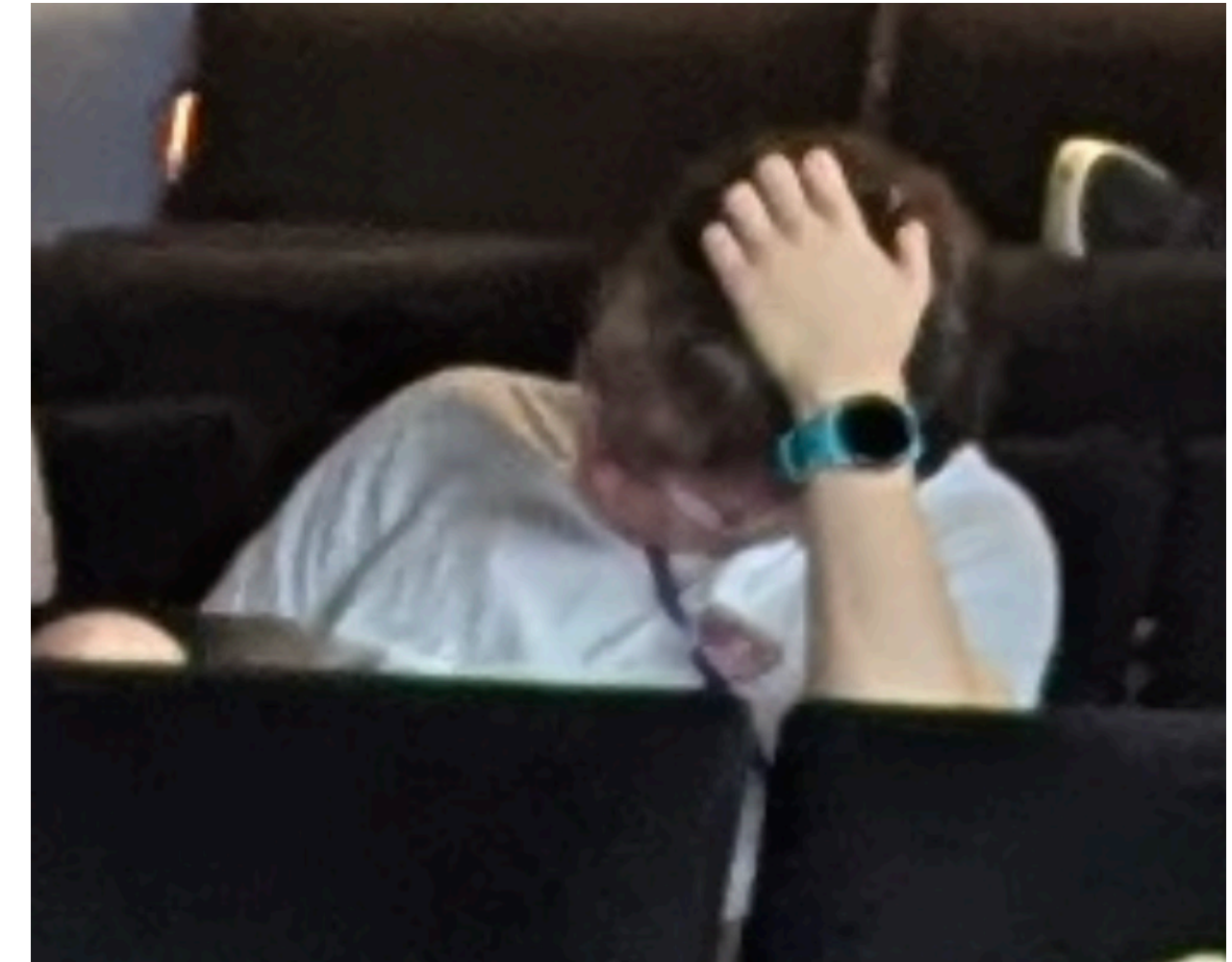






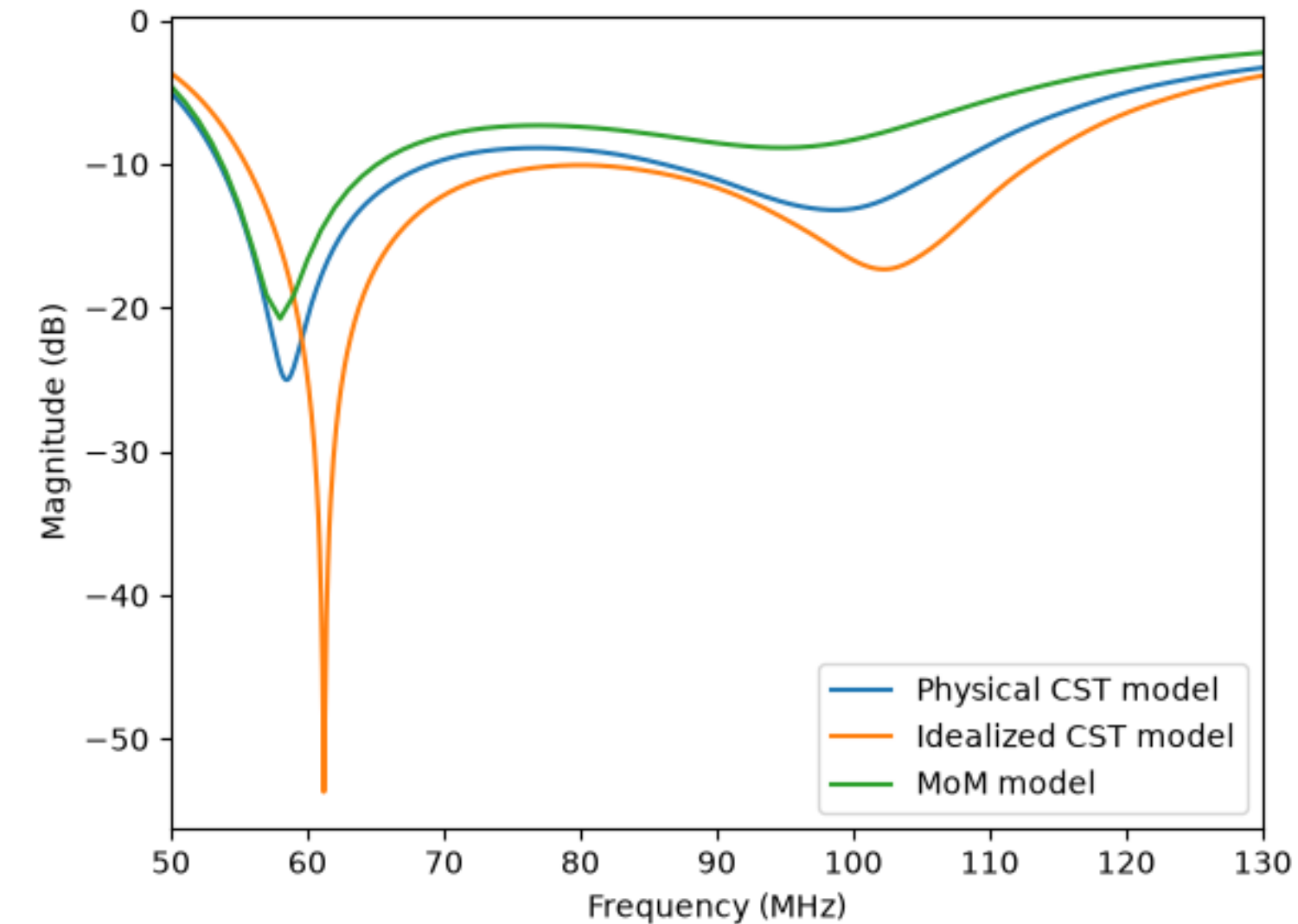
Future Plans - Paper

- Working to integrate this in to the pipeline
- The normalisation of the simulations was updated recently -> emulator needs retraining
- Need to be a bit more careful with testing, training and validation data
- Not expecting the results to change significantly
- Run some similar tests with the same example beams from Tutt et al. 2026

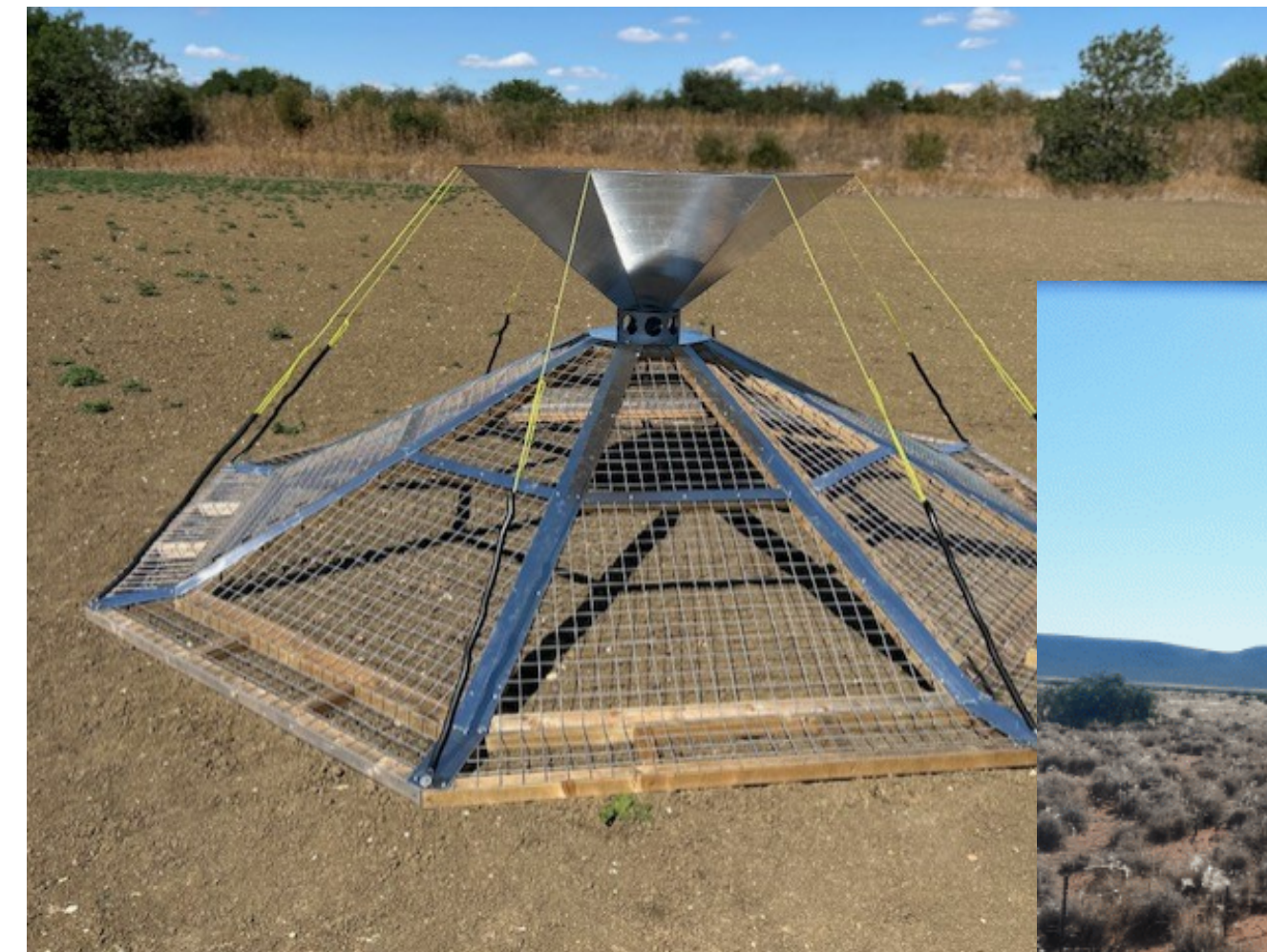


Future Plans - Dipole modelling and the Monopole

- Just submitted another DiRAC application
- Increase the number of dipole simulations for better accuracy
- Improve the dipole modelling
- Model and emulate the monopole
- Train the emulators
- Test the performance on mock observations
- Apply to real data



Credit: John Cumner



Conclusions

- In practice we can model the antenna gain pattern
- Still a bit of testing to do in the pipeline
- And a paper to write
- It's expensive and we are still limited by the accuracy of our simulation
- See Jacob's talk for an alternative way to account for the beams uncertainty

